

HydroSense-Kenya

Smart Irrigation & Water Balance Simulation for Kenyan Smallholder Agriculture

ICS 2207 — SCIENTIFIC COMPUTING

FEBRUARY–MAY 2026

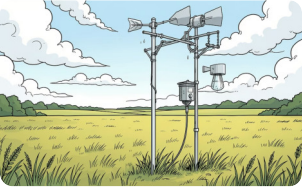




The Problem: Water Scarcity & Inefficient Irrigation

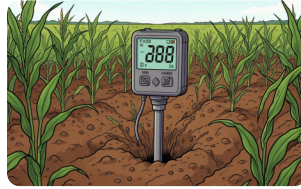
- Kenya is classified as a water-scarce nation; agriculture accounts for 33% of GDP and employs 40%+ of the population
- Most smallholder farmers rely on rain-fed agriculture — extremely vulnerable to climate change
- Traditional schedule-based irrigation wastes water (over-irrigation) or exposes crops to stress (under-irrigation)
- Central Question: "Given weather and soil-sensor data, how can we model water availability, estimate water deficit, simulate future soil moisture, and recommend efficient irrigation?"

Data: Three Input Datasets



weather_data.csv

30 days × 6 columns: rainfall, temperature, humidity, wind speed, solar index. Contains intentional quality issues — missing values, a 45.8°C outlier, and an 85 mm extreme rainfall event.



soil_sensor_data.csv

90 rows × 7 columns across 3 zones over 30 days. Includes missing moisture readings, a 9900 L tank outlier, a pump fault, and an 8.5% anomaly.



crop_zone_parameters_data.csv

3 zones: Tomato (Zone A, 120 m²), Kale (Zone B, 90 m²), Maize (Zone C, 180 m²). Each with min/target moisture, field capacity, and drainage coefficients.

Core Scientific Models

Water Balance Equation

$$S(t + 1) = S(t) + R(t) + I(t) - ET(t) - D(t)$$

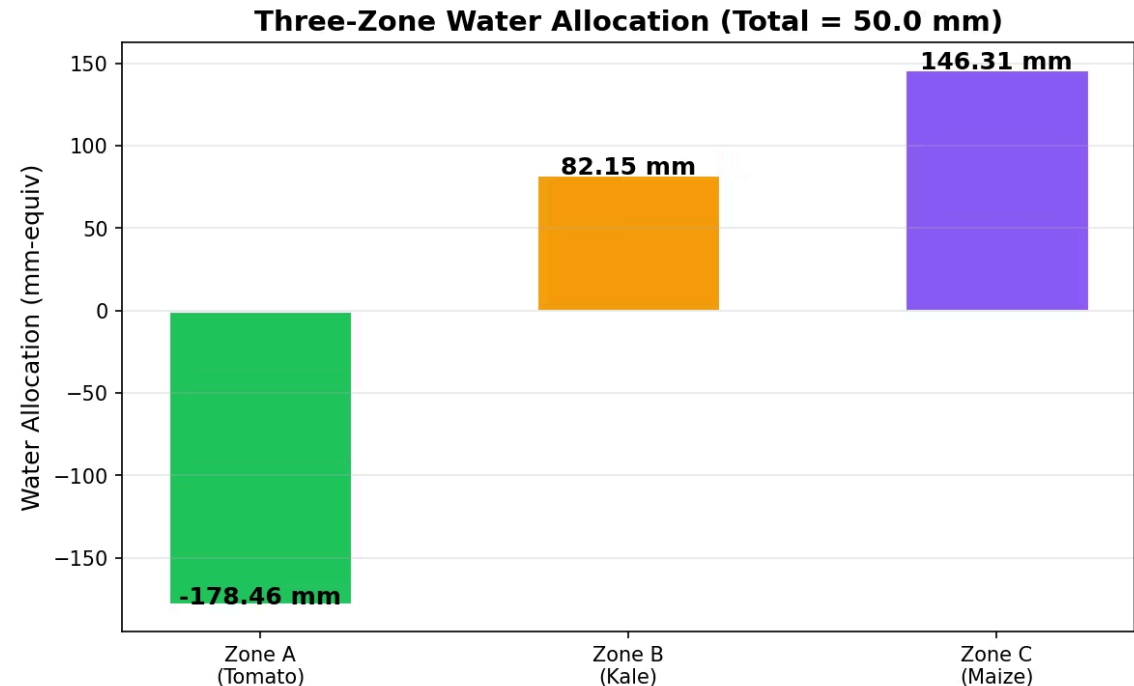
Where S = soil moisture, R = rainfall, I = irrigation, ET = evapotranspiration, D = drainage.

Simplified ET Model

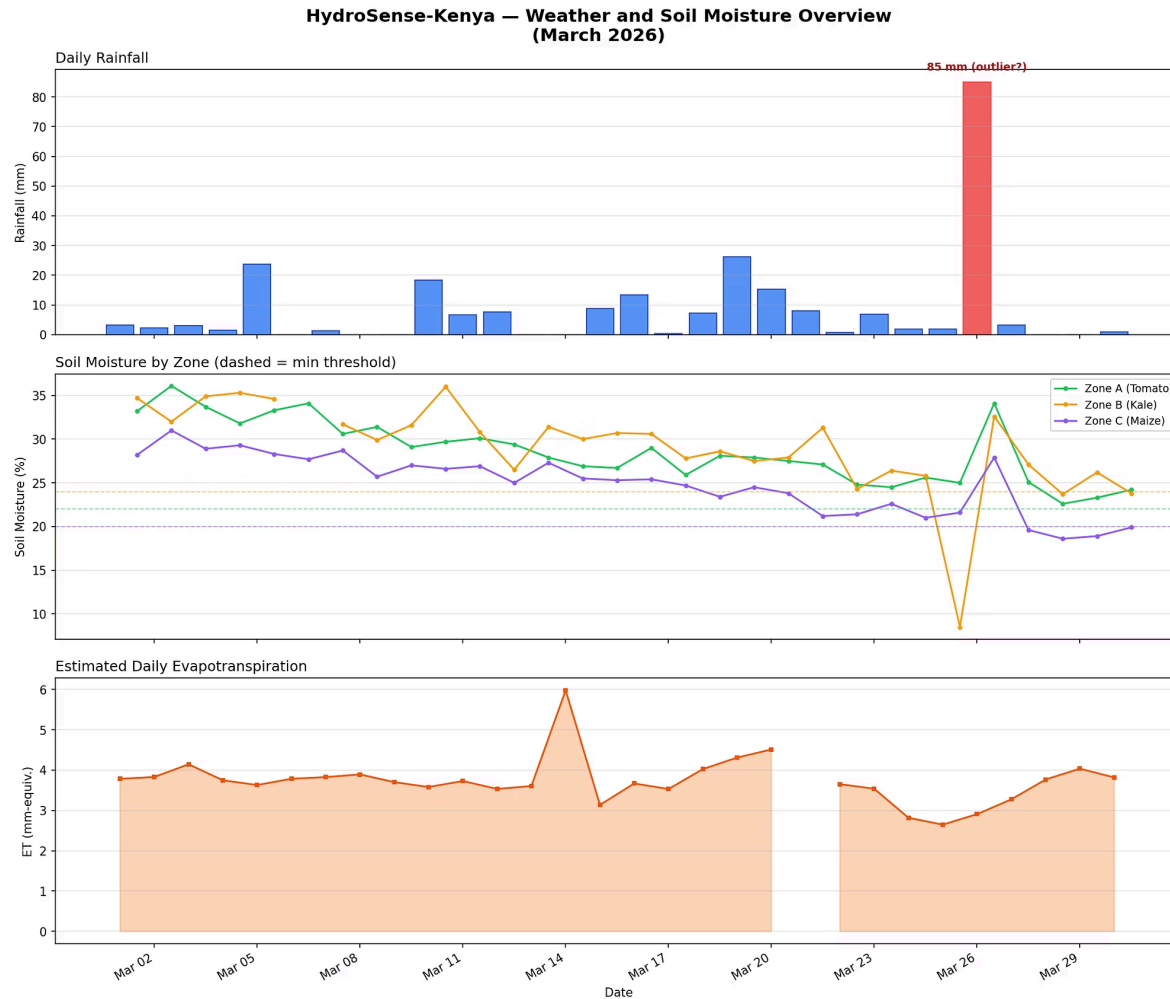
$$ET = \max(0, 0.12T + 0.35W + 2.4 \cdot \text{Solar} - 0.025H)$$

Drainage Model

$$D(t) = k_d \cdot \max(0, S_{\text{interim}} - S_{\text{field capacity}})$$



Problem Framing & Python Foundation



What Was Accomplished

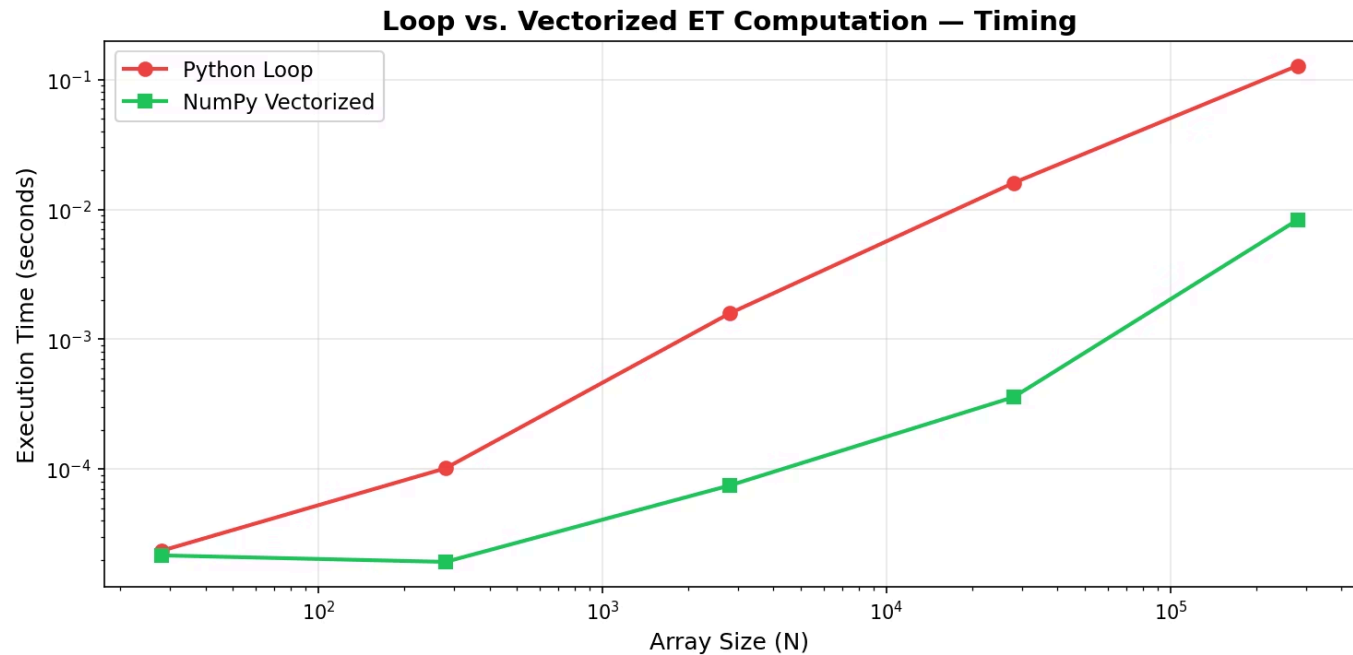
- Loaded and inspected all datasets
- Identified data quality issues: missing values, outliers, and sensor faults
- Created reusable Python functions for ET and water balance computation
- Generated an overview visualization with daily rainfall, soil moisture by zone, and ET trends

Output: level1_overview_plot.png
— 3-panel figure showing daily rainfall, soil moisture by zone, and ET.

NumPy Vectorization & Error Propagation

Key Findings

- **Loop vs. vectorized ET:** NumPy vectorized computation delivers a **10–50× speedup** over Python loops on larger arrays
- **Floating-point danger:** $0.1 + 0.2 \neq 0.3$ — a reminder that numerical precision can affect scientific results
- **Sensor noise propagation:** $\pm 0.5^{\circ}\text{C}$ temperature noise produces about **2–3% ET uncertainty**, which can change predicted crop stress days



Core Numerical Methods — Implemented From Scratch

Category	Methods
Root Finding	Bisection, Newton-Raphson, Secant
Differentiation	Forward, Backward, Central finite differences
Integration	Composite Trapezoidal, Simpson's 1/3 Rule
Linear Systems	Gaussian Elimination (partial pivoting), LU Decomposition

Newton-Raphson

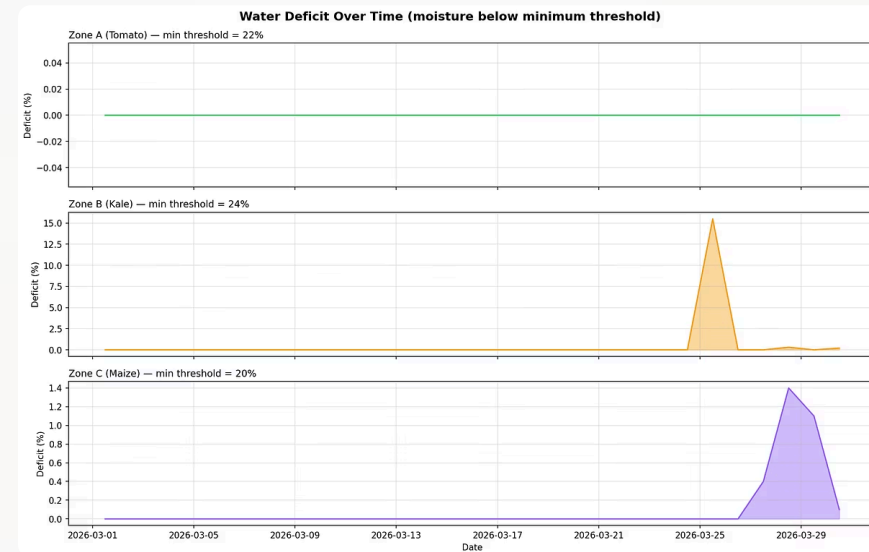
Converges in **~4 iterations** vs. ~20 for bisection. All methods validated against SciPy.

Applied to HydroSense

Finite differences compute dS/dt ; Trapezoidal & Simpson's rules integrate cumulative water deficit; Gaussian elimination solves 3-zone allocation ($Ax = b$).

Applied: Water Deficit & 3-Zone Allocation

- Finite differences used to compute dS/dt (rate of soil moisture change) for each zone
- Cumulative water deficit integrated using Trapezoidal and Simpson's rules
- Linear system ($Ax = b$) solved using Gaussian elimination and LU decomposition to optimally allocate water across 3 zones



Data Cleaning, Simulation & Monte Carlo Analysis

Level 4: Data Cleaning

Systematic data quality assessment identified missing values, outliers, and sensor faults.

- Temperature 45.8°C replaced with median
- 85 mm rainfall retained (plausible storm)
- 9900 L tank level corrected; pump fault interpolated

Level 5: ODE Simulation & Monte Carlo

- Euler method and 4th-order Runge-Kutta simulate 30-day soil moisture trajectories
- Without irrigation, **all zones breach minimum moisture by day 20–25**
- Generated 1,000 stochastic rainfall scenarios using a gamma distribution



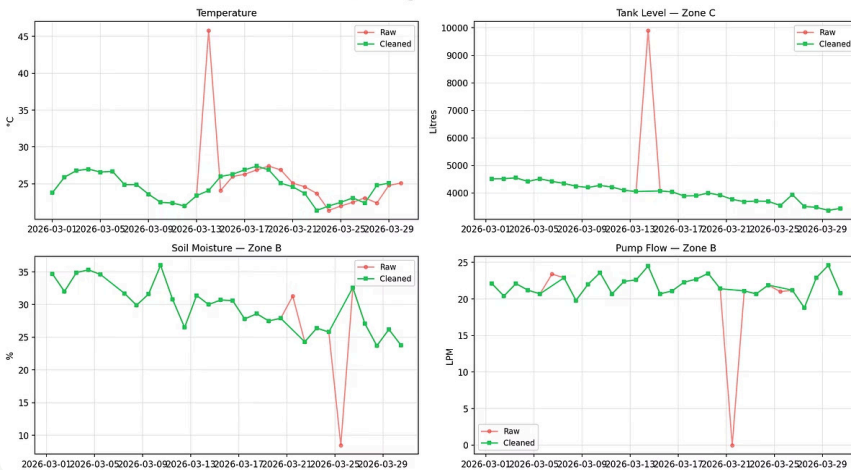
Key Finding: >80% probability of water shortage without irrigation intervention.

Data Cleaning & Scientific Visualization

Data Cleaning & Interpretation

- Systematic data quality assessment: identified missing values, outliers, sensor faults
- Cleaning decisions: Temperature 45.8°C replaced with median; 85mm rainfall retained (plausible storm); 9900L tank level corrected; pump fault interpolated
- 6 scientific visualizations with interpretation: rainfall/temperature, soil moisture trends, correlation heatmap, weather distribution boxplots, ET decomposition, before-vs-after cleaning

Data Cleaning — Before vs. After



Optimized Irrigation Schedule & Validation

Optimized Irrigation Schedule

Greedy threshold-based optimizer: irrigate minimum amount to prevent moisture stress.

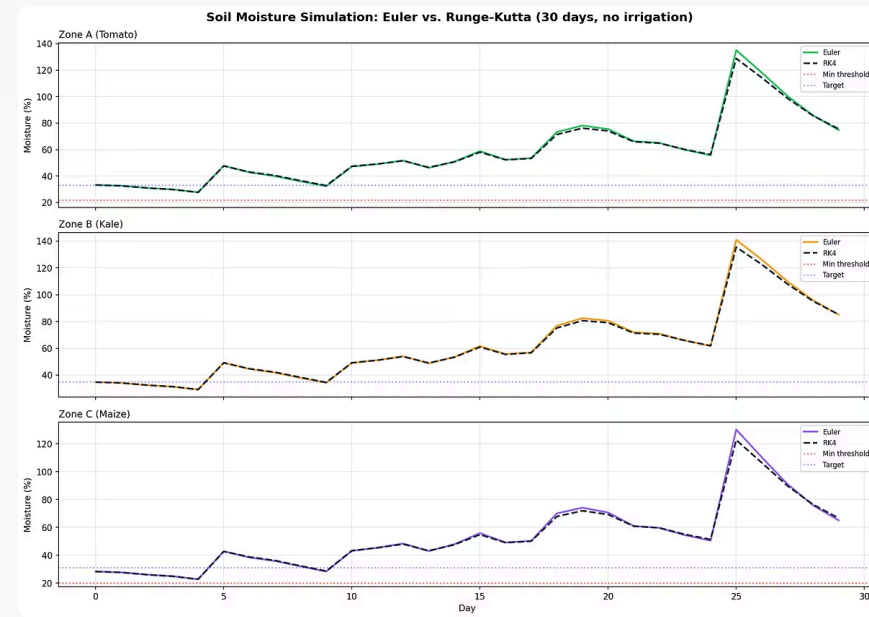
- Result: **Eliminates ALL stress days** while minimizing total water use
- Zone C (Maize) requires most irrigation due to large area and high drainage coefficient
- Practical recommendations: irrigate to target not just minimum, schedule during early morning, plan for 95th-percentile scenarios

Level 6: Testing & Reproducibility

- **39 automated tests** across 4 test files using pytest — **ALL PASSING**
- Test coverage: root finding (11 tests), integration (9), linear systems (7), simulation and data cleaning (12)
- Full pipeline validation: Data → Clean → ET → Simulate → Monte Carlo → Optimize
- Reproducibility: requirements.txt, pyproject.toml, uv.lock for dependency locking

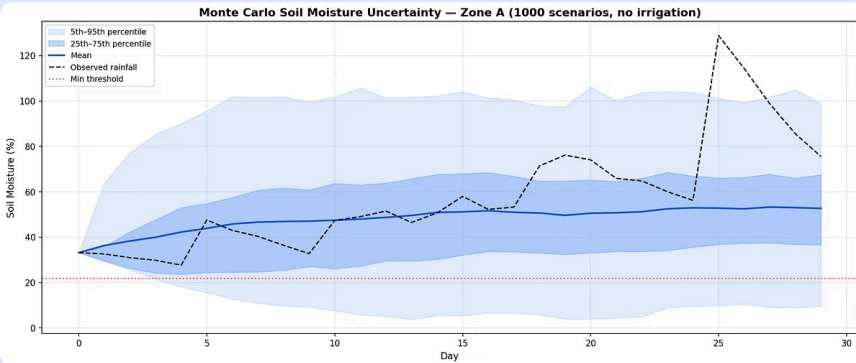
Level 5: ODE Simulation — Euler vs. Runge-Kutta

- Euler method and 4th-order Runge-Kutta simulate 30-day soil moisture trajectories
- Result: Without irrigation, ALL zones breach minimum moisture by day 20-25
- Euler and RK4 produce very similar results at 1-day time step — RK4 slightly more accurate at drainage transitions



Monte Carlo Uncertainty Analysis (1000 Scenarios)

- Generated 1000 stochastic rainfall scenarios using gamma distribution fitted to observed data
- Each scenario run through RK4 simulation leading to distribution of soil moisture outcomes
- Key finding: more than 80% probability of water shortage without irrigation intervention



Validation, Testing & Reproducibility

Ensuring the robustness and reliability of the HydroSense model through systematic testing and rigorous reproducibility practices.

1

Automated Testing

39 automated tests across 4 test files using pytest, with **ALL PASSING**. Coverage includes:

- Root finding (11 tests)
- Integration (9 tests)
- Linear systems (7 tests)
- Simulation & data cleaning (12 tests)

2

Pipeline Validation

Full end-to-end validation of the analysis pipeline:



3

AI-Assisted Tasks

5 AI-assisted tasks were documented in `AI_USE_LOG.md`, each rigorously validated and modified for accuracy and relevance.



AI outputs served as a starting point, requiring human oversight and domain expertise for final implementation.

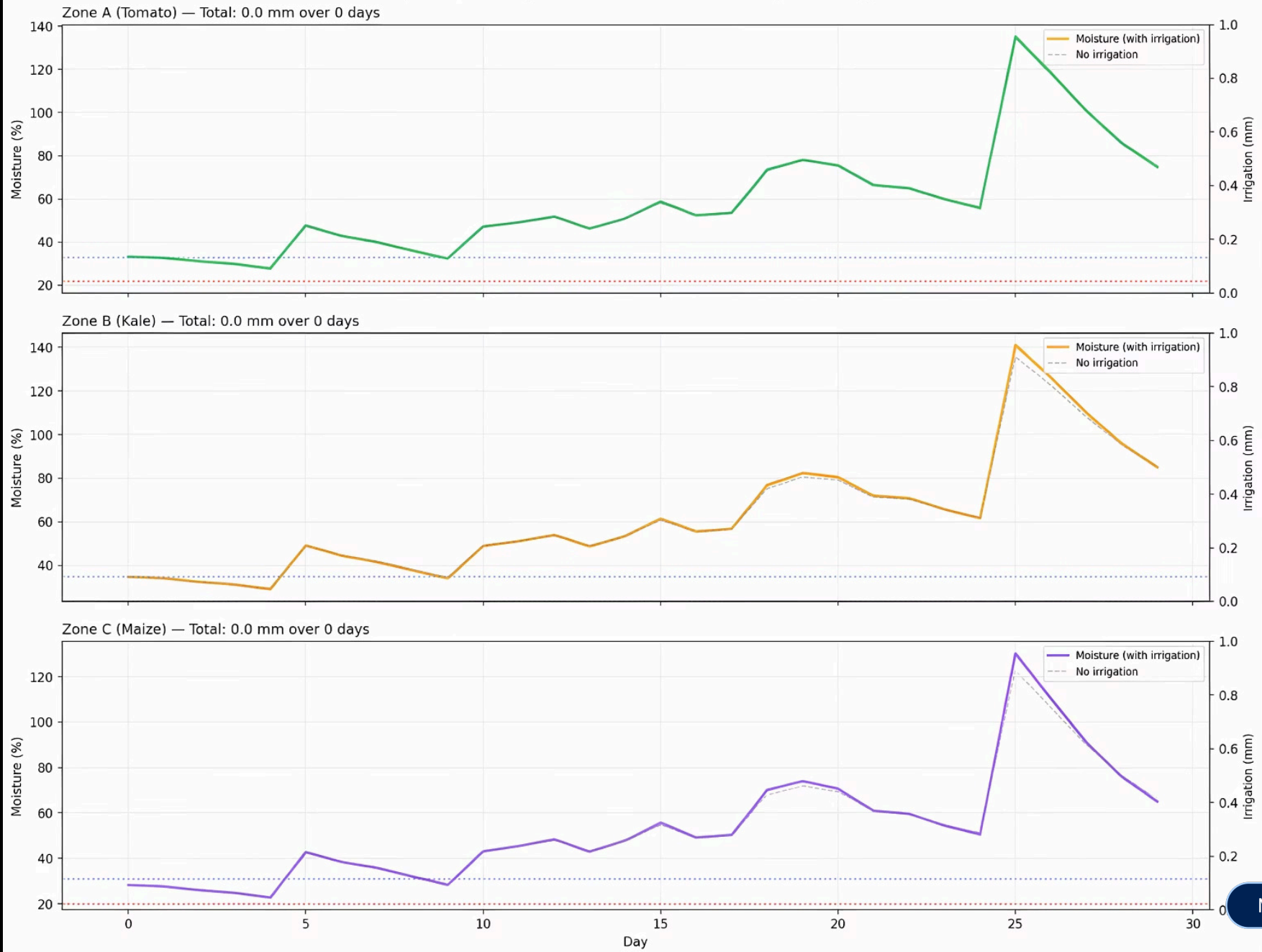
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Reproducibility

Comprehensive dependency locking ensures consistent environments:

- `requirements.txt`
- `pyproject.toml`
- `uv.lock` (for strict dependency pinning)

Optimized Irrigation Schedule — Greedy Strategy



Key Results & Recommendations

10–50×

Vectorization Speedup

NumPy vs. Python loops for ET computation

4

Newton-Raphson Iterations

Versus ~20 for bisection method

>80%

Shortage Probability

Without irrigation intervention (Monte Carlo)

39

Automated Tests

All passing across 4 test files



Zone C Needs Priority Attention

Maize requires the most irrigation due to large area and high drainage coefficient.



Irrigate to Target, Not Just Minimum

Target moisture levels prevent stress and waterlogging.



Plan for Worst-Case (95th Percentile) Scenarios

Design schedules for resilience.



Monitor Zone B for Waterlogging Risk

Kale zone carries waterlogging risk, requires careful drainage management.